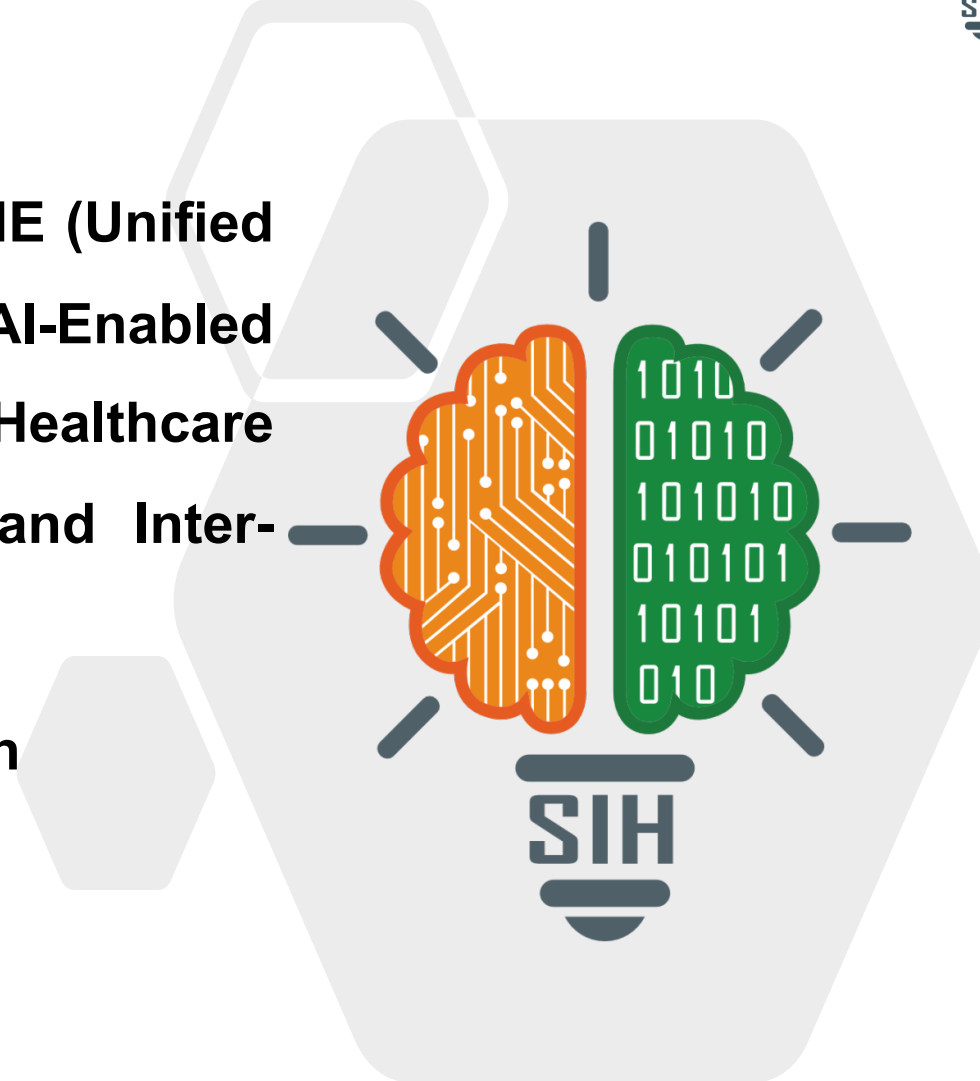


SMART INDIA HACKATHON 2026



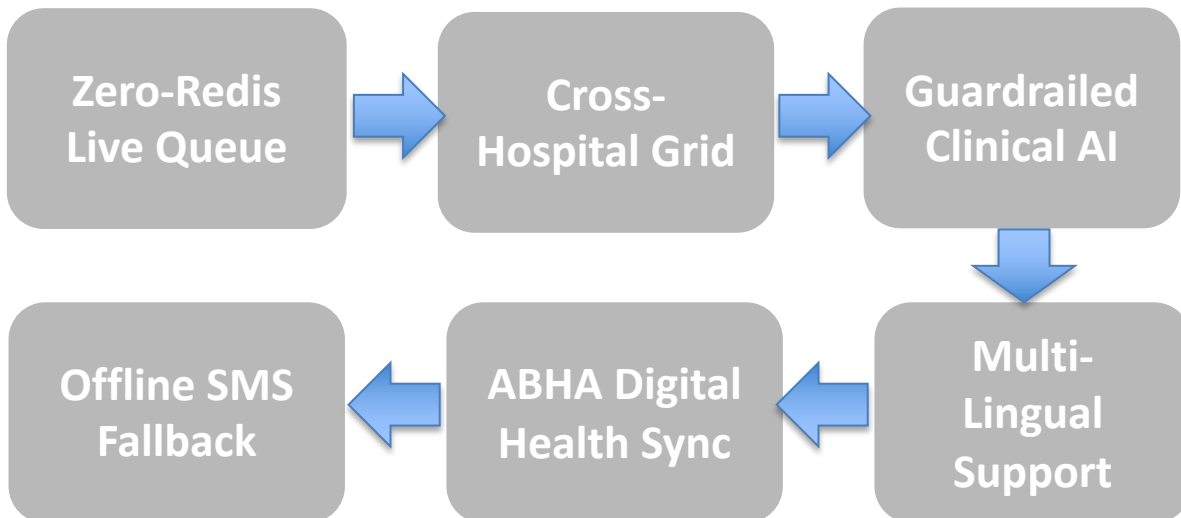
- **Problem Statement ID – 26198**
- **Problem Statement Title- Upchaar-UDHE (Unified Digital Healthcare Ecosystem): An AI-Enabled Real-Time Platform for Intelligent Healthcare Coordination, Resource Availability, and Inter-Hospital Connectivity**
- **Theme- MedTech / BioTech / HealthTech**
- **PS Category- Software**
- **Team ID- NA**
- **Team Name - TechX**



❖ Addressing the Problem –

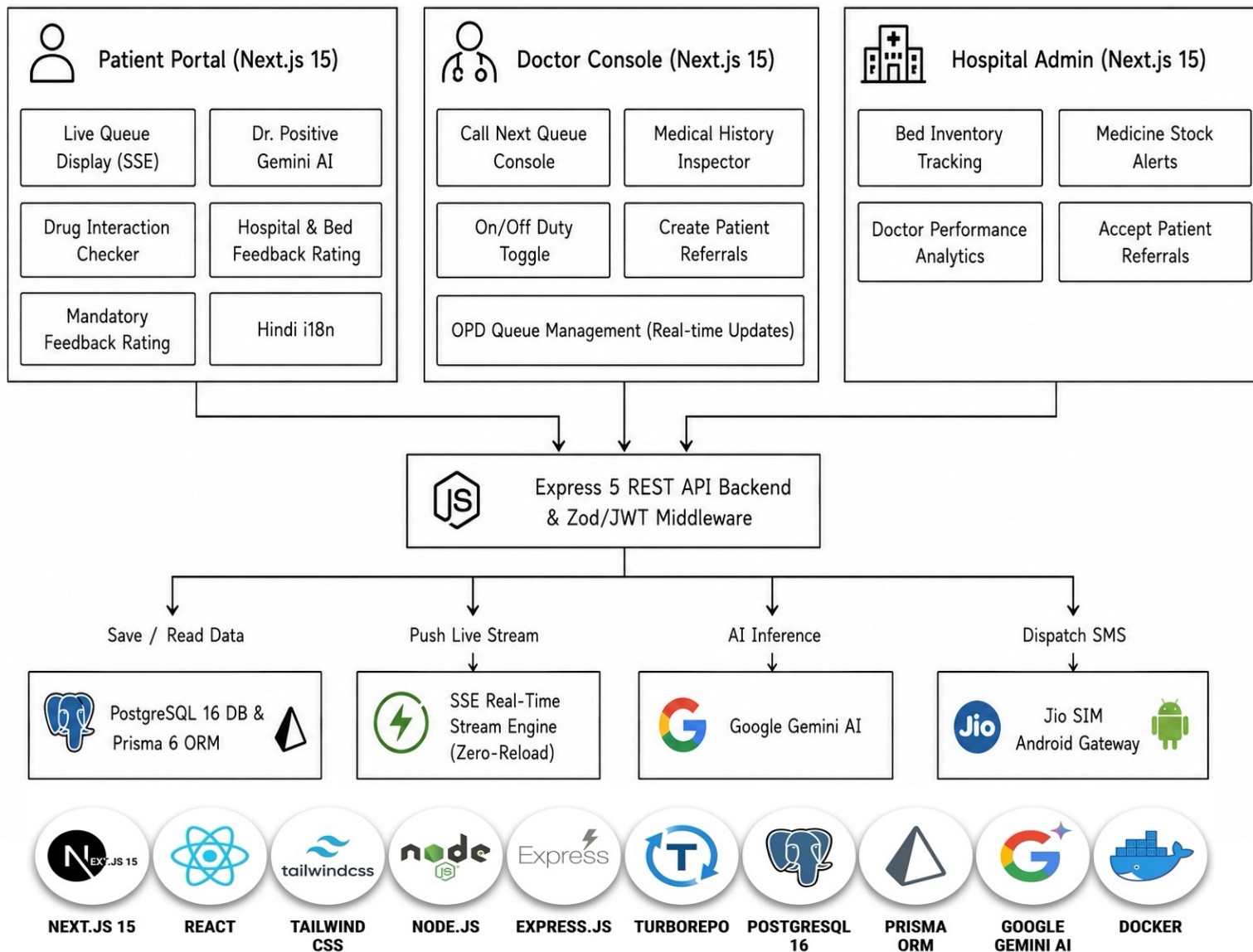
- **Overcrowded OPDs:** Patients endure **3–5 hours of blind waiting**.
- **Zero Live Visibility:** No real-time tracking for **queues or doctors**.
- **Fragmented Hospitals:** No cross-hospital sync for **emergency beds & blood**.
- **Paper-Based Delays:** Manual paperwork causes **critical referral bottlenecks**.

❖ Innovation and Uniqueness –



❖ Proposed Solution –

- **Unified Multi-Role Workspace:** Synchronized, access-scoped portals for **Patients, Doctors, and Hospital Admins**.
- **Real-Time Live OPD Queue:** **SSE-driven** queue tracking with **live queue position, people ahead, and estimated wait times**.
- **Smart Hospital Resource Console:** Dynamic tracking of **ICU & general beds** with automated **low-stock pharmacy alerts**.
- **AI Medical & Diagnostic Assistant:** Integrated **Dr. Positive (Gemini AI counsellor)** and **drug interaction report generator**.
- **Smart ABHA Digital Health Stack:** Native integration with national **Ayushman Bharat Digital Mission (ABDM)** for unified, interoperable patient health IDs.
- **Emergency Cross-Hospital Grid:** Inter-hospital , **live blood bank tracking, EHR patient referrals and medicine sharing**.



❖ Technologies Used -

- **Frontend Framework:** Next.js 15, React, and Tailwind CSS for responsive, multi-role web apps.
- **Backend & Monorepo:** Node.js & Express REST API structured inside a Turborepo monorepo.
- **Database & ORM:** PostgreSQL 16 database managed with Prisma ORM for type-safe queries.
- **Real-Time Engine:** Server-Sent Events (SSE) for zero-reload, live push queue updates.
- **AI Integration:** Google Gemini API powering Dr. Positive counselor and drug interaction checks.
- **Security & Validation** JWT Authentication paired with strict Zod schema validation.

❖ Feasibility -

- Built on Next.js, Express, PostgreSQL, and Prisma — **production-ready** from day one.
- **One backend** powers all three portals — Patient, Doctor, and Admin.
- Uses SSE for instant live queue updates — **no extra infrastructure needed or Redis needed**.
- Gemini AI runs via API — **zero GPU or hosting cost**.
- Scales to any hospital with **zero custom code changes**.
- **JWT + Zod** secures and validates every API request automatically.
- Modular monorepo structure makes **adding new features or portals fast and independent**.

❖ Key Challenges and risks -

- Handling 100s of concurrent OPD bookings without data conflicts.
- AI responses risk being mistaken for **clinical diagnosis**.
- Doctor resistance to replacing paper-based queue systems.
- Keeping AI drug data updated as new medicines enter the market.
- Rural areas face **poor internet connectivity**.

❖ Viability /Mitigation -

- **Optimistic UI + queue locking** prevents concurrent booking conflicts.
- AI always shows **"consult your doctor" disclaimer** on every response.
- **Simple one-click interface** designed for non-tech-savvy hospital staff.
- Drug data sourced from **verified medical APIs**, updated automatically.
- **SMS fallback alerts** keep patients informed even when fully offline.

Impact

- **PATIENTS**
 - Trusted doctor discovery
 - Easy appointment & queue management
 - AI-Powered Drug Interaction Checker & Dr. Positive Counsellor
 - Better access to healthcare information
 - Smart SMS Notification
- **DOCTORS**
 - Instant Pre-Consultation EHR
 - One-Click "Call Next" Console
 - Reduced Clinical Burnout
 - Smart Patient Referral System
- **HOSPITALS**
 - Smart Pharmacy & Expiry Alerts
 - Cross-Hospital Emergency Grid
 - Real-Time Bed Occupancy Console

Benefits

- **SOCIAL IMPACT**
 - Better healthcare accessibility
 - Improved Patient–Doctor–Hospital coordination
 - Easier understanding of medicine information
 - Enables faster access to the right specialist and improves continuity of care.
- **ECONOMIC IMPACT**
 - Reduced administrative workload
 - Better utilization of healthcare resources
 - Less dependency on fragmented/manual processes
 - Reduces unnecessary consultations and repeated tests.
- **ENVIRONMENTAL IMPACT**
 - Reduced paper-based records
 - Minimized Pharmaceutical Waste
 - Reduced Carbon Footprint & Travel Emissions

Research

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